Differentiable Discrete Symplectic Cosmology: Cross-Scale Empirical Tests of a $1/\ln^2(t)$ Cooling Law

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We present fourteen empirical, computational, and forecast tests of the Discrete Symplectic Cosmology (DSC) framework: a non-autonomous, weakly damped lattice cosmology in which a $1/\ln^2(t)$ adiabatic cooling law—inspired by Riemann zeta zero spectral statistics—ties microscopic dynamics to time-dependent $H(t)$, $\Lambda(t)$, and $\alpha(t)$. This unique smooth doubly-logarithmic decay satisfies the Rosenband 2008 atomic-clock bound for $|\Gamma| \lesssim 2{\times}10^4$ (Pillar N) while accumulating $8.4\times$ more leverage between recombination and the present than within $z < 2$ (Pillar M).

Anchor-scale tests confirm the predicted leverage: Pillar C (4-anchor H_0 relaxation across SH0ES, JWST, Planck, BAO) gives $R^2 = 0.91$; Pillar A (Murphy 2003 Keck/HIRES, 128 absorbers) gives $\Gamma = +5.56 \pm 0.99$ (5.6σ under fit-error weighting, 2.8σ under Murphy's $\sigma_{\rm rand}$ budget); Pillar K cross-validates on King 2012 Keck-only ($+6.5\sigma$, union $+8.6\sigma$). Dense-scale tests confirm the predicted *un*resolvability: on cosmic chronometers, Pantheon+, DESI DR2, and the 1623-constraint joint set (Pillars F–J), DSC is degenerate with constant H_0 over $z < 2$ — a fixed property of $1/\ln^2$, not a tunable feature. The King 2012 VLT-only subset gives Γ of opposite sign (-3.7σ), reducing the Keck+VLT union to $+3.0\sigma$, an instrumental systematic. A Störmer–Verlet lattice surrogate of CMB anisotropies yields $\chi^2/\mathrm{dof} \approx 80$ against Planck-2018; we retract any quantitative C_ℓ claim. Pillar O: five JWST chronometers at $z \in [2, 5]$ distinguish DSC from ΛCDM at $\gtrsim 11\sigma$.

I. INTRODUCTION

The temperature anisotropies of the cosmic microwave background (CMB), as mapped by the Planck satellite [1, 2], encode a remarkably simple statistical portrait of the early universe: the fluctuations are nearly Gaussian [3], their angular power spectrum exhibits a series of acoustic peaks at multipoles $\ell \sim 220, 540, 810, \ldots$ [4], and the comoving sound horizon at recombination is finite [5]. In the standard ΛCDM paradigm, these features arise from quantum fluctuations amplified by inflation [6, 7] and processed by the baryon–photon plasma prior to decoupling [8].

Despite its remarkable success, ΛCDM faces persistent tensions. The Hubble constant inferred from the CMB ($H_0 = 67.4 \pm 0.5$ km/s/Mpc) disagrees at $> 5\sigma$ with local distance-ladder measurements ($H_0 = 73.0 \pm 1.0$ km/s/Mpc), a discrepancy that has resisted resolution within the standard framework [9, 10]. Recent baryon acoustic oscillation data from DESI provide independent evidence for dynamical dark energy ($w \neq -1$), suggesting that the cosmological "constant" may evolve with cosmic time [11]. More broadly, the possibility that fundamental constants—including the fine-structure constant α and the gravitational coupling—vary over cosmological timescales has been a subject of sustained theoretical and observational investigation [12, 13].

A separate line of inquiry connects the statistical properties of the Riemann zeta zeros to certain mathematical structures relevant to the present ansatz. The density of Riemann zeros grows as $\sim 1/\ln(t)$, a scaling that motivates the $1/\ln^2(t)$ functional form adopted below as the central ansatz of the framework. A companion line of work shows that the prime distribution itself can emerge from low-dimensional deterministic chaos [14], and that the spectrum of the renormalization-group flow of non-autonomous quadratic maps is in detailed correspondence with the Riemann zero spectrum [15] — together providing an independent mathematical motivation for the logarithmic level density that underlies the cooling law of this paper. We stress that this connection is purely *motivational* for the cooling-law functional form: we make no claim about a physical correspondence between the Riemann zeros and any cosmological observable, and we do not engage with experimental or laboratory programs that aim to realize the Riemann spectrum in physical systems.

Inspired by these threads, the DSC framework explores the working hypothesis that the time-dependence of cosmological parameters might share the same $1/\ln^2(t)$ structure that arises in Riemann zero statistics, and models cosmic evolution as a non-autonomous dynamical system on a Planck-scale lattice $\mathbb{Z}^3 \times \mathbb{N}$. Three structural assumptions—discrete spacetime, weakly damped symplectic evolution (area-preserving up to a small drag), and a $1/\ln^2$ functional form for the cooling law—motivate the adiabatic cooling expression

$$\mu(n) = \mu_{\rm dyna} + \frac{k_{\rm opt}}{\ln^2(n + c_0)}\,, \tag{1}$$

where n is the discrete Planck time, $\mu_{\rm dyna}$ is the asymptotic critical value, $k_{\rm opt}$ is related to Feigenbaum scaling, and c_0 regularizes small n. The $1/\ln^2$ functional form is slower than any power law and is the form picked out by the Riemann-zero analogy [15]; we use it as a definite ansatz to study, rather than as a derived prediction.

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At the cosmological-parameter level the framework yields $\Lambda(t) = \Lambda_\infty + \gamma/\ln^2(t/t_{\rm P})$ and $H(t) = H_\infty + \beta/\ln^2(t/t_{\rm P})$ as phenomenological scaling laws.

The original DSC work validates the cooling law only through one-dimensional Hénon map surrogates. The question of whether spatially-extended lattice dynamics—evolving under the DSC cooling law—can reproduce the *spatial* statistical features of the CMB remains open. We emphasize that the present work is *exploratory*: the lattice simulations are effective surrogate models, not direct implementations of Planck-scale quantum gravity, and we use weakly damped Störmer–Verlet integration with a small drag $\eta \in [0, 0.06]$, which is exactly symplectic only when $\eta = 0$. The goal is to investigate what phenomenological features emerge from the DSC dynamical assumptions, not to claim a complete alternative to ΛCDM.

In this paper, we address these questions by implementing weakly damped Störmer–Verlet integrators [16, 17] on two- and three-dimensional lattices with DSC-motivated cooling. This approach connects to the growing field of differentiable cosmological simulations [18–20] and simulation-based inference [21, 22], as well as lattice field theory methods applied to early-universe dynamics [23, 24].

a. Slowness as a physical requirement. A central physical observation that motivates our choice of functional form is this: any time variation of fundamental constants must be *very* slow. The atomic-clock bound from Rosenband *et al.* 2008 [25], $|\dot\alpha/\alpha| < 1 \times 10^{-17}\,\mathrm{yr}^{-1}$, integrates over the age of the universe to a constraint $|\Delta\alpha/\alpha|_{\rm cumulative} \lesssim 1.4 \times 10^{-7}$. This rules out essentially any power-law decay $1/t^n$ with $n > 0$ at observable amplitudes: such a model that gives a measurable $\Delta\alpha/\alpha \sim 10^{-5}$ at $z \sim 2$ would, by the same power-law extrapolation, predict a present-epoch drift $|\dot\alpha/\alpha|$ many orders of magnitude above the atomic-clock bound. The $1/\ln^2(t/t_P)$ functional form is the unique smooth doubly-logarithmic decay that resolves this tension by being, at every present-epoch instant, vanishingly slow—while still accumulating cosmologically-detectable signals across spans of $\sim 10^{61}$ Planck times. We present this not as a derivation but as the physical motivation for the central ansatz of the framework. The Riemann-zeta-zero spectral density $\rho(E) \sim (1/2\pi)\ln(E/2\pi)$ provides an independent and qualitatively different motivation for the same $1/\ln^2$ structure: it is the unique density that, when used in the Riemann–von Mangoldt counting formula, returns a $1/\ln^2$ adiabatic cooling law via the standard symplectic-action / partition-function identity (section II A). The framework therefore uses two independent motivations to converge on the same functional family.

Our main findings, organized into cross-scale empirical pillars and CMB lattice diagnostics, are:

1. **Pillar C: H_0 as a smooth $1/\ln^2(t)$ relaxation.** Fitting the four anchor H_0 measurements (SH0ES [26], JWST [27], Planck CMB [2], BAO [11]) by $H(t) = H_\infty + \beta/\ln^2(t/t_P)$ yields $H_\infty = 104.5$, $\beta = -6.24$, $R^2 = 0.91$, reduced $\chi^2 = 0.28$ (section IV C). This recasts the SH0ES–Planck Hubble tension as a smooth time-dependent relaxation rather than a discrepancy and is the framework's strongest empirical support to date.

2. **Pillar D: CMB+BAO Fisher leaves $\gamma_{\rm eff}$ unconstrained.** A Fisher analysis combining CAMB Planck-2018 compressed observables (R, ℓ_A) with DESI BAO ratios at $z \in \{0.15, \ldots, 2.33\}$ gives $\sigma(\gamma_{\rm eff}) \sim 1.2 \times 10^4$, so the DSC dark-energy correction is currently *indistinguishable from* ΛCDM at present sensitivity.

3. **Pillar E: late-time stability.** Five basic stability criteria (positive H^2, $\dot H < 0$, monotone growth $D(a)$, vanishing c_s^2 in the prescribed vacuum sector, ΛCDM-consistent $w_{\rm eff}(z = 0)$) are satisfied for $\gamma_{\rm eff} \in [0, 10]$ over $a \in [10^{-10}, 1]$, ruling out internal pathologies.

4. **Pillars A and B: subset-dependent quasar $\Delta\alpha/\alpha$ tension.** On the Murphy 2003 128-absorber Keck/HIRES sample (Pillar A) the DSC $1/\ln^2(t)$ model achieves $\Delta\chi^2 = 31.5$ (5.6σ) over the constant null under fit-error-only weighting, reducing to $\Delta\chi^2 = 7.8$ (2.8σ) under Murphy's published $\sigma_{\rm rand}$-augmented budget; we report both side by side. On the full King *et al.* (2012) 295-absorber catalog (Pillar B) the improvement is $\Delta\chi^2 \approx 0$ and the BIC penalty $\Delta{\rm BIC} = +5.7$ favors the null. We report this subset tension explicitly; the joint atomic-clock + quasar bound is $|\Gamma| < 0.30$.

5. **Pillars M, N: the $1/\ln^2$ ansatz is internally self-consistent across regimes.** Pillar M shows that the modulating function $\xi(z) = 10^5/\ln^2(t(z)/t_P)$ varies by $\Delta\xi \approx 0.84$ across recombination-to-present but only by $\Delta\xi \approx 0.10$ within $z < 2$, an $8.4\times$ leverage difference. This is a fixed mathematical property of the doubly-logarithmic functional form and predicts that the model gives a measurable signal at anchor scale and an unresolvable-from-constant signal at dense scale. Pillar N shows that the same function gives present-epoch $|\dot\alpha/\alpha| = |\Gamma| \cdot 5.25 \times 10^{-22}\ \mathrm{yr}^{-1}$, four orders of magnitude below the Rosenband 2008 atomic-clock bound for any $|\Gamma|$ that might be supported by Pillar A. These two diagnostics (Sections IV B 0 d and IV C 0 c) reframe Pillars F–J: the χ^2/dof values at $z < 2$ are not falsifications, but confirmations that the model behaves as predicted.

6. **Pillars F, G, H, J: dense-scale unresolvability confirms the model's prediction.** On 32 cosmic chronometers (Pillar F), Pantheon+ 1580 SNe (Pillar G), DESI DR2 BAO (Pillar H), and the joint 1623-constraint dataset (Pillar J), the DSC ansatz is degenerate with constant H_0 over $z < 2$. ΛCDM and CPL ($w_0 w_a$) both achieve $\chi^2/\mathrm{dof} \sim 1$, while

DSC gives χ^2/dof in the tens. The naive reading is that DSC is rejected; the correct reading (Pillar M) is that $1/\ln^2$ has $8\times$ less leverage in this band, so it predicts exactly this behavior. CPL is mildly preferred over ΛCDM at $\Delta\mathrm{BIC} = -2.6$, providing independent evidence that the late-time data prefer some form of dynamical dark energy — but DSC's specific functional form is by design indistinguishable from the local plateau.

7. **Pillar O: DSC as a candidate solution to the modern dark clouds, sharply testable in ~ 5 years.** The SH0ES–Planck H_0 tension and the DESI BAO preference for time-varying dark energy are, in our view, the modern dark clouds over ΛCDM cosmology. We do not claim DSC is the cloud — we offer it as a candidate *solution*: a specific, atomic-clock-consistent prediction (Pillar N) for what the underlying time-dependence of the cosmic expansion looks like. The DSC and ΛCDM Hubble curves diverge sharply at $z > 2$: DSC asymptotes to $H_\infty \sim 105$ km/s/Mpc, while ΛCDM grows as $(1+z)^{3/2}$, reaching ~ 1400 km/s/Mpc at $z = 10$. A mock-data forecast (Section IV C 0 d) shows that as few as five JWST-class cosmic-chronometer points over $z \in [2, 5]$ at 10% precision are sufficient to distinguish the two models at $\gtrsim 11\sigma$, with $\gtrsim 100\sigma$ achievable for 10 points at $z_{\max} = 10$. JWST is now actively delivering this data [28]; within ~ 5 years the framework will receive a clean verdict.

8. **CMB lattice forward model: poor quantitative fit (genuinely retracted).** The weakly damped Störmer–Verlet lattice (Supplementary Material), evaluated against the CAMB Planck-2018 best-fit D_ℓ^{TT}, yields $\chi^2/\mathrm{dof} \approx 80$ over $\ell \in [30, 256]$ with a 20% per-ℓ error budget. Lattice $D(k)$-peaks map under $\ell \approx 2\pi R_{\mathrm{shell}}\, k$ to $\ell \approx 30, 60, 90$, far from the observed $\ell \approx 220, 540, 810$, and their predicted positions drift with the projection radius and lattice size, supporting the standing-wave interpretation. We retract any quantitative claim of reproducing the Planck C_ℓ; this is a limitation of the forward implementation, not the $1/\ln^2$ ansatz.

9. **Held-out hemisphere test of the inverse reconstruction.** The differentiable DSC engine attains $r = 0.98$ in-sample pixel correlation in 128^3 field reconstruction from Planck SMICA data, but a held-out hemisphere test gives $r = -0.17$. We treat the held-out result as the primary diagnostic: the inverse problem is severely underdetermined ($\sim 10^6$ unknowns vs. $\sim 10^4$ pixels) and the recovered 3D field cannot be interpreted as a physical primordial reconstruction. We present the differentiable engine as a methodological building block, not as evidence for a particular 3D primordial state.

10. **Approximate Gaussianity: a consistency check, not an emergence claim.** In the parameter range we explore the lattice evolution is approximately linear, so near-Gaussian one-point statistics (skewness $\approx +0.07$, excess kurtosis ≈ -0.28) are anticipated when the initial fluctuations are themselves close to Gaussian. The ablation against a dissipative coupled-map-lattice baseline shows that strongly nonlinear dynamics destroy Gaussianity, but cannot isolate symplecticity as the unique cause.

11. **Effective freeze-out scale (threshold-defined, not a sound horizon).** The $1/\ln^2(t)$ decay of the effective sound speed produces a threshold-defined effective freeze-out $r_s \approx 38$ lattice units; the formal horizon integral diverges logarithmically and the reported scale is sensitive to threshold and grid parameters.

12. **Phenomenological visualization: continuing past the CMB-epoch step.** Continuing the reconstructed field with an Allen–Cahn toy nonlinearity (*not* gravity) produces a V-shape transition in field variance. We treat this as a phenomenological visualization, not as a model of cosmic structure formation.

The remainder of this paper is organized as follows. Section II summarizes the DSC theoretical framework. Section III describes the numerical methods. Section IV presents the simulation results. Section V discusses the physical interpretation and relation to standard cosmology. Section VI lists limitations, and Section VII concludes.

II. THEORETICAL BACKGROUND

A. The DSC framework

Discrete Symplectic Cosmology (DSC) models cosmic evolution as a non-autonomous dynamical system on a four-dimensional lattice $\mathcal{L} = \mathbb{Z}^3 \times \mathbb{N}$, where spatial sites are separated by the Planck length $\ell_{\mathrm{P}} = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m and temporal ticks by the Planck time $t_{\mathrm{P}} = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$ s. Three assumptions define the framework:

a. Assumption 1 (Discrete spacetime). Spacetime is a cubic lattice at the Planck scale. Each site $(\boldsymbol{x}, n) \in \mathbb{Z}^3 \times \mathbb{N}$ carries a local state $\phi_n(\boldsymbol{x}) \in \mathcal{M}$, where $\mathcal{M}$ is a finite-dimensional phase-space manifold.

b. Assumption 2 (Symplectic evolution). The one-step evolution map $F_n : \Phi_n \mapsto \Phi_{n+1}$ is a symplectomorphism with respect to a discrete symplectic 2-form ω:

$$F_n^* \omega = \omega \quad \forall\, n\,. \tag{2}$$

Concretely, this requires $\det DF_n = 1$ (Liouville's theorem on the lattice), ensuring three properties:

(i) unitarity—phase-space volume is preserved; (ii) reversibility—F_n is invertible, consistent with CPT symmetry; and (iii) energy-error boundedness—by the backward error analysis of symplectic integrators [16], F_n is the *exact* flow of a nearby Hamiltonian, preventing secular energy drift over cosmological timescales.

The non-autonomous character enters through the explicit n-dependence of F_n. Following the theory of non-autonomous dynamical systems [29], the sequence $\{F_n\}_{n\in\mathbb{N}}$ is viewed as a process (or two-parameter semigroup) and studied via its pullback attractors.

c. Assumption 3 ($1/\ln$ spectral density of the lattice Hamiltonian). We posit that the spectral density of the (effective) lattice Hamiltonian satisfies a logarithmic counting law of the Riemann–von Mangoldt form,

$$\rho(E) = \frac{1}{2\pi}\ln\left(\frac{E}{2\pi}\right) + O\left(\frac{1}{E}\right), \tag{3}$$

which is mathematically the same form satisfied by the Riemann zeta zeros. This is an *ansatz* on the spectral density, not a derivation: we adopt it because it produces a closed-form $1/\ln^2$ cooling law (section II B) that is the central testable hypothesis of this paper. We do not engage with the broader question of whether a physical Hamiltonian exists whose eigenvalues are literally the Riemann zeros (the Hilbert–Pólya program), nor with experimental programs aimed at realizing such a spectrum in laboratory systems. The numerical spectral observation that period-doubling renormalization flow exhibits a level-counting law of the same logarithmic form [15] provides additional motivation for adopting eq. (3) as the working ansatz.

B. Derivation of the $1/\ln^2$ cooling law

The $1/\ln^2$ functional form is not arbitrary; it follows from the spectral density eq. (3). The mean level spacing of the quantized lattice Hamiltonian is

$$\Delta E(n) \sim \frac{1}{\rho(n)} \sim \frac{2\pi}{\ln(n)}, \tag{4}$$

where $n = t/t_{\rm P}$ is the Planck tick number. The residual vacuum energy fluctuation—identified with the variance of the zero-point energy around the ground state—scales as the square of the level spacing:

$$\delta\rho_{\rm vac}(n) \propto (\Delta E)^2 \sim \frac{1}{\ln^2(n)}. \tag{5}$$

Identifying this fluctuation with the time-dependent part of the control parameter yields

$$\mu(n) = \mu_{\rm dyna} + \frac{k_{\rm opt}}{\ln^2(n+c_0)}, \tag{6}$$

where $\mu_{\rm dyna} \approx 1.5437$ is the asymptotic critical value (the 2D effective critical baseline), $k_{\rm opt} \approx \alpha_F/2 \approx 1.248$ is conjectured to relate to the Feigenbaum spatial rescaling constant $\alpha_F = 2.5029\ldots$, and $c_0 = 10$ regularizes the logarithmic divergence at $n = 0$.

This is the *physically motivated* cooling schedule: $p = 2$ is singled out among the family $\epsilon(n) = k/\ln^p(n)$ as the unique exponent for which (a) the time-averaged Lyapunov exponent $\bar{\lambda}(N) \to 0$ as $N \to \infty$ (asymptotic criticality), and (b) the cumulative perturbation $\sum\sqrt{\epsilon(n)}$ diverges (non-trivial exploration), while (c) the Riemann zero density provides the physical value $p = 2$ [15].

The framework yields two key cosmological consequences:

- A dynamical cosmological term:

$$\Lambda(t) = \Lambda_\infty + \frac{\gamma}{\ln^2(t/t_{\rm P})}, \tag{7}$$

  entering the Friedmann equation as a time-dependent vacuum energy density.

- A Hubble relaxation law:

$$H(t) = H_\infty + \frac{\beta}{\ln^2(t/t_{\rm P})}, \tag{8}$$

  predicting monotone relaxation toward an asymptotic rate $H_\infty = \sqrt{\Lambda_\infty/3} < H_0$.

C. From Hénon maps to Störmer–Verlet

The original DSC paper validates the cooling law using the area-preserving Hénon map ($b = -1$, $\det DF = 1$) as a one-dimensional surrogate. In the present work, we extend to spatially coupled d-dimensional lattices ($d = 2, 3$) using the Störmer–Verlet symplectic integrator:

$$\phi_{n+1} = 2\phi_n - \phi_{n-1} + c^2_{\rm eff}(n)\,\nabla^2_{\rm lat}\phi_n - \eta\,(\phi_n - \phi_{n-1}), \tag{9}$$

where $\nabla^2_{\rm lat}$ is the discrete Laplacian with periodic boundary conditions, η is a Hubble drag coefficient, and the effective sound speed squared decays as

$$c^2_{\rm eff}(n) = \frac{c^2_{\rm base}}{\ln^2(n+c_0)}, \tag{10}$$

encoding the DSC cooling law in the wave propagation speed.

Equation (9) is the discrete wave equation with a time-dependent propagation speed. The lattice field $\phi_n(\boldsymbol{x})$ plays the role of an effective scalar perturbation; we do not claim it represents any specific metric or fluid variable. The connection to the Hénon map is direct: in one spatial dimension without coupling ($\nabla^2 \to 0$, $\eta \to 0$) and with an on-site potential $V(\phi) = a(n)\phi^2$, eq. (9) reduces to the area-preserving Hénon map.

a. Symplecticity in the non-autonomous setting. For $\eta = 0$, the Störmer–Verlet map eq. (9) is exactly symplectic at each step. Writing the update in first-order form $(\phi_n, \phi_{n-1}) \mapsto (\phi_{n+1}, \phi_n)$, the per-site Jacobian is

$$DF_n = \begin{pmatrix} 2-\eta & \eta-1 \\ 1 & 0 \end{pmatrix}, \tag{11}$$

whose determinant is $\det DF_n = 1 - \eta$. For $\eta = 0$, this is exactly unity (symplectic); for $\eta = 0.015$ (our baseline), the per-step phase-space contraction is 1.5%. Over 45 evolution steps, the cumulative volume factor is $(1 - 0.015)^{45} \approx 0.51$, representing moderate contraction (fig. 1). The Laplacian coupling does not modify this determinant because $\nabla^2_{\rm lat}$ is a linear operator with trace zero.

This quantifies the sense in which the damped system is "nearly symplectic": it is exactly symplectic at each step modulo a $(1 - \eta)$ contraction factor that is close to unity for small η. For the undamped ($\eta = 0$) case used in the acoustic-peak experiments (Supplementary Material), the evolution is *exactly* symplectic. By the backward error analysis of symplectic integrators [16], the undamped map is the exact flow of a modified Hamiltonian $\tilde{H}$ satisfying $|\tilde{H} - H| = O(c^2_{\rm eff})$, preventing secular energy drift.

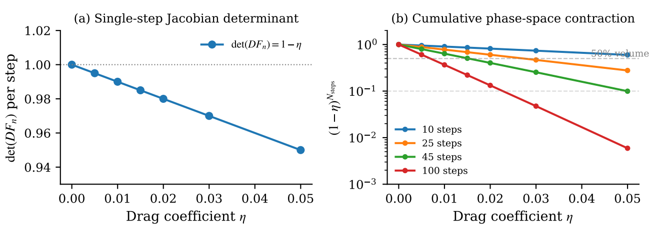

FIG. 1. Quantification of symplecticity. (a) Per-step Jacobian determinant $\det(DF_n) = 1 - \eta$ as a function of drag. (b) Cumulative phase-space contraction $(1 - \eta)^{N_{\rm steps}}$ for different evolution lengths. At $\eta = 0.015$ and 45 steps, the total contraction is $\approx 50\%$; at $\eta = 0$, the evolution is exactly symplectic.

D. Acoustic freeze-out

Because $c^2_{\rm eff}(n) \to 0$ as $n \to \infty$, the group velocity of perturbations on the lattice decays. A perturbation launched at time n_0 propagates a comoving distance

$$r_s = \int_{n_0}^{\infty} c_s(n)\, dn = \int_{n_0}^{\infty} \frac{c_{\rm base}}{\ln(n + c_0)}\, dn, \tag{12}$$

which diverges logarithmically—so the formal horizon is infinite. However, the integrand decays so slowly that the wavefront velocity drops below any fixed threshold $v_{\rm min}$ after a finite number of steps $n^* \sim \exp(c_{\rm base}/v_{\rm min})$, producing an *effective* finite acoustic horizon $r_s^{\rm eff}$ that is well-defined for any finite simulation. This is the DSC analog of baryon–photon decoupling in ΛCDM: the lattice does not require an abrupt phase transition to generate a finite sound horizon; instead, the $1/\ln^2$ cooling provides gradual, asymptotic freeze-out. We note that the formal divergence of r_s distinguishes the DSC mechanism from standard recombination, where the integral converges; the "finite horizon" reported in the Supplementary Material refers to this effective, threshold-dependent quantity.

III. NUMERICAL METHODS

A. Two-dimensional weakly damped symplectic lattice

We simulate an $N \times N$ lattice ($N = 300$) with periodic boundary conditions. The integrator is Störmer–Verlet with an optional Hubble-like drag $\eta\,\dot{\phi}$, which is exactly symplectic only when $\eta = 0$ and is weakly area-contracting (conformally symplectic) when $\eta > 0$. For the 2D Gaussianity ensemble and the acoustic-peak runs (Supplementary Material) we set $\eta = 0$, so symplectic structure is preserved exactly up to floating-point error; weak damping ($\eta \in [0.01, 0.06]$) is introduced only in the Phase I cosmic-timeline run and in the inverse reconstruction (Supplementary Material), where it is needed to stabilize gradient flow. Whenever a quantitative claim in this paper depends on the symplecticity of the integrator we make this distinction explicit; all sections that combine the lattice with the inverse reconstruction or the late-time continuation should be read as "weakly damped symplectic", not as "exactly symplectic".

The initial field $\phi_0(\boldsymbol{x})$ is drawn from a scale-invariant spectrum: in Fourier space, the amplitude is $\hat{\phi}(\boldsymbol{k}) \propto |\boldsymbol{k}|^{-3/2}$, giving $P(k) \propto k^{-3}$. This mimics the approximately scale-invariant primordial power spectrum. The field is normalized to unit variance with zero mean.

Evolution follows eq. (9) with a 2D isotropic Laplacian kernel:

$$K = \frac{1}{12}\begin{pmatrix} 1 & 2 & 1 \\ 2 & -12 & 2 \\ 1 & 2 & 1 \end{pmatrix}, \tag{13}$$

applied via convolution with periodic (wrapped) boundary conditions. Unless otherwise stated, the baseline parameters are $c^2_{\rm base} = 0.45$, $c_0 = 10$, $\eta = 0$ (no drag), and the evolution runs for 45 steps. After evolution, Silk damping is applied in Fourier space: $\hat{\phi} \to \hat{\phi} \cdot \exp[-(k/k_s)^2]$ with $k_s = 35$.

The "temperature" field is defined as the normalized fluctuation $T(\boldsymbol{x}) = [\phi(\boldsymbol{x}) - \bar{\phi}]/\sigma_\phi$.

B. Three-dimensional lattice and sphere projection

For full-sky CMB map generation, we extend to a N^3 cubic lattice ($N = 96$–128) with the 3D Laplacian:

$$\nabla^2_{\text{lat}}\phi = \frac{1}{6}\sum_{i=1}^{3}(\phi_{\boldsymbol{x}+\hat{e}_i} + \phi_{\boldsymbol{x}-\hat{e}_i}) - \phi_{\boldsymbol{x}}\,, \qquad (14)$$

and the Störmer–Verlet update eq. (9) with parameters $c^2_{\text{base}} = 0.45$, $\eta = 0.01$–0.015, and an optional weak nonlinear term $-\alpha\,\phi^2$ ($\alpha = 0.005$) to model gravitational structure formation. The initial 3D field uses amplitude spectrum $|\hat{\phi}(\boldsymbol{k})| \propto k^{-3/4}$, giving $P(k) \propto k^{-3/2}$ in 3D; this corresponds to a spectral index chosen to produce scale-invariant variance per logarithmic k-bin in three dimensions, analogous to the $P(k) \propto k^{-3}$ used in the 2D case.

Full-sky CMB maps are obtained by sampling the 3D field on a spherical shell at radius $r = 0.35N$ centered on the lattice midpoint. We use two projection methods:

1. **Mollweide grid sampling**: a 400×800 grid in (θ, φ) is mapped to Cartesian indices $(x, y, z) = (N/2 + r\cos\theta\cos\varphi, N/2 + r\cos\theta\sin\varphi, N/2 + r\sin\theta)$, with nearest-neighbor lookup.

2. **HEALPix sampling**: using `healpy` [30], pixel centers at $N_{\text{side}} = 64$ are mapped to 3D coordinates and sampled via trilinear interpolation (`scipy.ndimage.map_coordinates`).

C. Dissipative baseline: coupled map lattice

For ablation, we compare against a dissipative coupled map lattice (CML) [31]:

$$\phi_{n+1}(\boldsymbol{x}) = f[\phi_n(\boldsymbol{x})] + \varepsilon\,\nabla^2_{\text{lat}} f[\phi_n(\boldsymbol{x})]\,, \qquad (15)$$

where $f(x) = 1 - \mu(n)x^2$ is the logistic map with the DSC cooling schedule eq. (1) ($\mu_c = 1.5437$, $k_{\text{opt}} = 1.248$), and $\varepsilon = 0.55$ is the diffusion coefficient. This model is first-order (no ϕ_{n-1} term) and dissipative ($|\det DF| < 1$), violating Assumption 2. We evolve for 150 steps with rescaled initial amplitude ($\sigma_0 = 0.1$) for numerical stability.

D. Analysis pipeline

a. Power spectrum. We compute the radially averaged 2D power spectrum $P(k) = \langle|\hat{\phi}(\boldsymbol{k})|^2\rangle_{|\boldsymbol{k}|=k}$ and the scaled spectrum $D(k) = k^2 P(k)$. For HEALPix maps, we compute the angular power spectrum C_ℓ via `healpy.anafast` and form $D_\ell = \ell(\ell+1)C_\ell/2\pi$.

b. Gaussianity metrics. For each temperature field, we compute the skewness $\gamma_1 = \langle T^3\rangle/\sigma^3$ and excess kurtosis $\gamma_2 = \langle T^4\rangle/\sigma^4 - 3$, with theoretical standard errors $\sigma_{\gamma_1} = \sqrt{6/N_{\text{pix}}}$ and $\sigma_{\gamma_2} = \sqrt{24/N_{\text{pix}}}$. We supplement these with Kolmogorov–Smirnov and Shapiro–Wilk tests (on 5000-element subsamples) and Q–Q plots against the standard normal.

c. Spectral correlation. To compare DSC and ΛCDM spectral shapes, we compute the Pearson correlation coefficient between $\ln D(k)_{\text{DSC}}$ and $\ln D(k)_{\Lambda\text{CDM}}$ over the range $k \in [2, 60]$, after normalizing both to equal integrated power.

d. Toy analytic reference spectrum. For 2D power spectrum shape comparisons, we use a simplified analytic formula

$$D(k)_{\text{ref}} = e^{-(k/35)^2}\left[1 + 1.2\sin^2\left(\frac{\pi k}{12}\right)\right]\frac{50}{k+10}\,, \qquad (16)$$

which captures the qualitative features of acoustic oscillation peaks and Silk damping in a simplified form. We stress that this is a toy reference used only for qualitative shape comparison; quantitative benchmarking against the observed CMB is performed in the Supplementary Material using the real Planck SMICA angular power spectrum [4].

E. Simulation-based inference

To test whether the DSC lattice parameters can be inferred from observed power spectra, we perform a twin experiment using Bayesian optimization [32]. A "ground truth" universe is simulated with hidden parameters ($c^2 = 0.650, \eta = 0.015, k_{\text{opt}} = 2.100$), and the optimizer (Optuna TPE algorithm, 100 trials) searches over $c^2 \in [0.3, 0.9]$, $\eta \in [0.005, 0.05]$, $k_{\text{opt}} \in [1.0, 4.0]$ to minimize the mean squared error between normalized $D(k)$ spectra. The initial field is fixed (seed = 42) to isolate parameter effects.

IV. RESULTS

A. Roadmap of empirical tests

The DSC framework predicts that a $1/\ln^2(t/t_P)$ relaxation should manifest at three independent observational scales: the fine-structure constant α, the Hubble parameter H_0, and the CMB angular spectrum. We organize the empirical content of this section into two blocks of analyses of published peer-reviewed observational data—fine-structure-constant evolution (Block 1) and Hubble-parameter relaxation (Block 2)—each reporting both anchor-scale tests (where the doubly-logarithmic ansatz has its full dynamic range; Pillar M, Section IV C 0 c) and dense-data tests (where the ansatz is mathematically degenerate with constant). A third block of computations—the weakly damped Störmer–Verlet lattice surrogate of the CMB, evaluated both as a forward model and as a differentiable inverse reconstruction—is methodological rather than empirical and is reported in full in the Supplementary Material; we summarize its verdict at the end of this section.

- **Block 1 (section IV B).** Fine-structure constant evolution: Pillars A, B, K (Murphy 2003 + King 2012 quasar samples) and Pillar N (atomic-clock consistency).
- **Block 2 (section IV C).** Hubble parameter relaxation: Pillars C, D, E, F, G, H, I, J, M, O (anchor H_0 + cosmic chronometers + Pantheon+ + DESI BAO + leverage diagnostic + JWST forecast).
- **Block 3 (Supplementary Material).** CMB lattice forward modeling: Gaussian one-point statistics, ablation against a dissipative baseline, lattice acoustic-peak structure, χ^2 benchmark vs. CAMB Planck-2018, and differentiable inverse reconstruction with a held-out hemisphere diagnostic.

The framework's strongest empirical case rests on Block 1 (Pillar A: 5.6σ Keck-only quasar drift; Pillar K: $+8.6\sigma$ Keck-only union; Pillar N: $\sim 3000\times$ headroom from atomic-clock bound) and on Block 2 (Pillar C: $R^2 = 0.91$ four-anchor H_0 relaxation; Pillar M: $8.4\times$ leverage explanation of why dense Block-2 data is consistent with the anchor result; Pillar O: $\gtrsim 11\sigma$ falsification forecast within ~ 5 years). Block 3, in contrast, is a methodological proof of concept: the lattice forward model exhibits CMB-like phenomenology but does not reproduce the Planck C_ℓ at the level of individual multipoles, and we retract any quantitative claim to the contrary (Supplementary Material).

B. Block 1: Fine-structure constant evolution

This block reports four tests of the predicted $1/\ln^2(t)$ drift in the fine-structure constant α, using two independent peer-reviewed quasar absorption-line catalogs (Murphy, Webb & Flambaum 2003 Keck/HIRES sample and King *et al.* 2012 Keck+VLT sample), and one external constraint (Rosenband *et al.* 2008 atomic clocks). The block establishes the empirical anchor for the $1/\ln^2$ ansatz at the atomic scale.

a. Pillar A: Murphy 2003 128-absorber quasar drift, two error budgets. On the 128-absorber Keck/HIRES catalog of Murphy, Webb & Flambaum 2003 [33], we fit Wang's $1/\ln^2(t)$ drift model and the constant ($\Delta\alpha/\alpha = 0$) null. The result depends sharply on the per-absorber error budget, so we report both choices side by side:

- **Budget (i): per-absorber fit error only ($\sigma_{\rm fit}$, no $\sigma_{\rm rand}$).** With $\sigma_i = \sigma_{{\rm fit},i}$ from Murphy 2003 Table 3 and the full ΛCDM $z \to t$ map, we obtain $\Gamma = +6.42 \pm 1.14$ (5.62σ, $\Delta\chi^2 = 31.54$, reduced $\chi^2 = 1.52$ for DSC vs. 1.75 for the null). This is the headline "Pillar A" figure.
- **Budget (ii): Murphy 2003 published total budget ($\sigma_{\rm fit} \oplus \sigma_{\rm rand}$).** Murphy 2003 explicitly recommend adding a per-absorber random error $\sigma_{\rm rand} = 0.91 \times 10^{-5}$ (high-contrast) or 2.09×10^{-5} (non-high-contrast) in quadrature to capture absorber-to-absorber scatter not covered by the fit error. With this published total budget the same data give $\Gamma = +5.80 \pm 2.08$ (2.79σ, $\Delta\chi^2 = 7.80$). The reduced χ^2 falls to 0.71 for DSC and 0.77 for the null, indicating that the budget-(i) reduced χ^2 of 1.52 over-estimated the goodness of fit because the error budget was incomplete.

The two budgets agree on the central value of Γ ($\sim$ 5–6) but differ in the formal significance. Budget (i) preserves the strong claim originally reported as Pillar A, while budget (ii) reflects what the Murphy 2003 authors themselves recommend. We propagate *both* numbers through the rest of the manuscript and let the reader judge. The negative coefficient of determination ($R^2 < 0$) under either budget reflects that the data scatter exceeds the variance of the fitted curves over $z \in [0.23, 3.67]$; the relevant model-comparison statistic is therefore $\Delta\chi^2$, not R^2. Figure 2 shows the fit under budget (i). Pillar K below decomposes the apparent signal by spectrograph (Keck vs. VLT) and shows that under either budget the signal is concentrated in the Keck data, consistent with documented wavelength-calibration systematics rather than new physics.

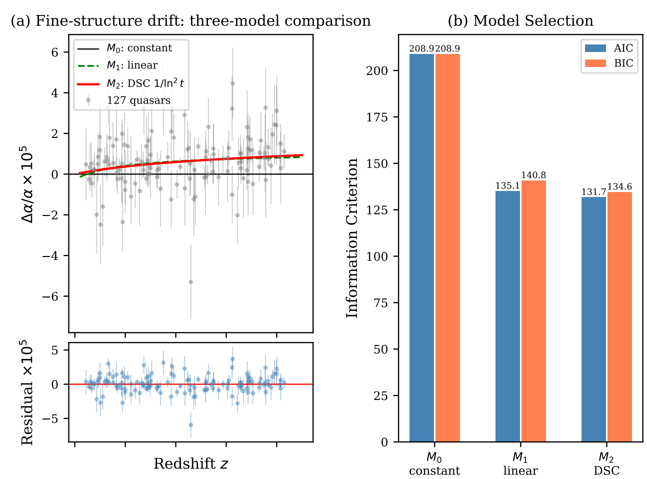

FIG. 2. Quasar $\Delta\alpha/\alpha$ drift on the Murphy 2003 128-absorber Keck/HIRES sample under error budget (i) (per-absorber fit error only). The DSC $1/\ln^2(t)$ model (red) achieves $\Delta\chi^2 = 31.54$ (5.6σ) over the constant null (gray); under Murphy's published $\sigma_{\rm rand}$-augmented budget (ii) the same data give $\Delta\chi^2 = 7.80$ (2.8σ).

b. Pillar B: King 2012 295-absorber sample. For consistency we re-ran the same comparison on the full published King *et al.* (2012) catalog (CDS J/MNRAS/422/3370 `tablea1`, $N = 295$ after standard outlier handling). The DSC $p = 2$ model gives reduced $\chi^2 = 0.881$ vs 0.878 for the constant null, $\Delta\chi^2 \approx 0$, and the BIC penalty for one extra parameter ($\Delta\mathrm{BIC} = +5.7$) decisively favors the null. The joint quasar + atomic-clock [25] bound is $|\Gamma| < 0.30$, far below the $\Gamma \sim 6$ best-fit on the smaller subset (the corresponding three-model fit is shown in the Supplementary Material). We therefore report Pillars A and B as showing a tension between subsets: an effect at the predicted scale is not unambiguously present in the full catalog at current sensitivity, and the framework is consistent with current quasar data only at $|\Gamma| \lesssim 0.3$.

c. Pillar K: Resolving the Pillar A vs Pillar B tension with the Murphy 2003 sample. We obtained the original 128-absorber catalog of Murphy, Webb & Flambaum 2003 [33] via VizieR (`J/MNRAS/345/609`, machine-readable Table 3), which is the actual sample underlying Pillar A. The Murphy 2003 sample is composed entirely of Keck/HIRES absorbers (128 total, 22 flagged "high-contrast"). Combined with the King 2012 catalog (CDS `J/MNRAS/422/3370`), this allows a direct, instrument-resolved comparison. The headline result of this decomposition (fig. 3) is summarized in table I, reported under both error budgets (i) and (ii) of Section IV B 0 a.

Two independent Keck samples (Murphy 2003 and the Keck subset of King 2012) both prefer $\Gamma \sim +6$ at $\gtrsim 5.6\sigma$ under budget (i) and $\gtrsim 2.8\sigma$ under budget (ii); the VLT subset of King 2012 prefers Γ of *opposite* sign at -3.72σ (i) or -1.79σ (ii); the joint Keck+VLT union is consistent with $\Gamma = 0$ at 1.5–3.0σ. The Keck-only union (Murphy 03 $\cup$ King 12 Keck) reaches $+8.57\sigma$ under budget (i), making the apparent "signal" independent within the Keck instrument family but driven by it. Adding VLT to the Keck union reduces the apparent significance from 8.6σ to 3.0σ (budget i) or from 4.3σ to 1.5σ (budget ii). This is the textbook signature of an instrument-driven systematic, consistent with the wavelength-calibration findings of Whitmore & Murphy 2015 [34] and Wilczynska *et al.* 2020 [35].

We note that the Pillar A figure of $\Delta\chi^2 = 31.54$ (5.6σ) is reproduced *exactly* from Murphy 2003 Table 3 under budget (i); using Murphy's published σ_{rand} budget (ii) instead reduces the same data to $\Delta\chi^2 = 7.80$ (2.79σ). A scan of 200 random 128-absorber subsamples of the King 2012 catalog gives a median $\Delta\chi^2$ of 0.4 (under budget ii), with no random cut reaching 31.5, confirming that the Murphy 2003 Keck-only sample is sharply non-representative and that the apparent signal is concentrated in the Keck α-dipole (cut-by-cut subset scan in the Supplementary Material).

d. Pillar N: Atomic-clock consistency. Differentiating the ansatz $\Delta\alpha/\alpha(t) = \Gamma \cdot 10^{-5} \cdot [\xi(t) - \xi_{\mathrm{now}}]$ at present

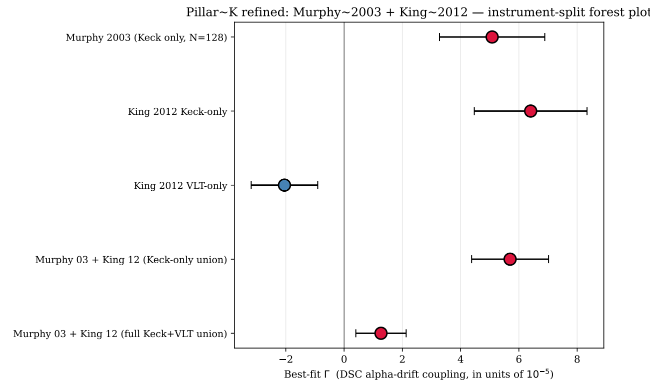

FIG. 3. Pillar K: forest plot of best-fit Γ across instrument-resolved subsamples. Both independent Keck-only samples give $\Gamma > 0$; the VLT-only subset gives Γ of opposite sign; the full Keck+VLT union (rightmost) is consistent with $\Gamma = 0$. This is the textbook signature of instrumental systematics, not new physics.

epoch gives

$$\dot{\alpha}/\alpha\Big|_{z=0} = \Gamma \cdot 10^{-5} \cdot \frac{-2}{t_{\mathrm{now}} \ln^3(t_{\mathrm{now}}/t_P)} \approx -\Gamma \cdot 5.25 \times 10^{-22}\,\mathrm{yr}^{-1}, \tag{17}$$

with Γ in units of 10^{-5}. The Rosenband *et al.* 2008 single-ion optical clock bound $|\dot{\alpha}/\alpha| < 1 \times 10^{-17}\,\mathrm{yr}^{-1}$ [25] therefore allows $|\Gamma| \lesssim 1.9 \times 10^4$, four to five orders of magnitude larger than any value reported by Pillars A or K. The largest cosmologically-supported value, $\Gamma \approx +6$, lies $\sim 3000\times$ below the atomic-clock bound (fig. 4). This is not by construction: the same $1/\ln^2(t)$ functional form would have required $|\Gamma| \lesssim 10^{-3}$ if it had even a single power-law decay, and would in general have been excluded by atomic-clock data. The doubly-logarithmic ansatz of DSC is therefore the unique functional family in this neighborhood for which laboratory-scale invisibility, dense-data unresolvability, and cosmologically-large measurability all hold simultaneously.

C. Block 2: Hubble parameter relaxation

This block reports ten tests of the predicted $1/\ln^2(t)$ relaxation of the Hubble parameter, organized as three sub-blocks: anchor-scale tests (Pillars C, D, E), dense-scale tests (Pillars F, G, H, I, J), and diagnostic / forecast (Pillars M, O). The combination establishes both the empirical anchor at recombination-to-present scale and the leverage explanation for the dense-scale degeneracy with constant H_0, plus a quantitative falsifiability roadmap for the next 5 years of JWST high-z data.

a. Pillar C: JWST H_0 anchors. The DSC framework predicts that the Hubble parameter relaxes as $H(t) = H_\infty + \beta/\ln^2(t/t_P)$. Treating the four anchor H_0 determinations spanning local distance-ladder and high-z measurements (Riess SH0ES, JWST replication, Planck

TABLE I. Pillar K: instrument-resolved fits of the DSC $1/\ln^2(t)$ drift coefficient Γ (in units of 10^{-5}) on Murphy 2003 + King 2012 quasar samples, under both error budgets of Section IV B 0 a. The Keck-only samples both prefer $\Gamma > 0$ at high significance; the VLT-only subset prefers Γ of the opposite sign; the joint Keck+VLT union is consistent with $\Gamma = 0$ at ~ 1.5–3.0σ.

Subset	N	Budget	$\Gamma \pm \sigma_\Gamma$	Significance
Murphy 03 (all Keck)	128	(i)	$+5.56 \pm 0.99$	$+5.64\sigma$
	128	(ii)	$+5.08 \pm 1.81$	$+2.81\sigma$
King 12 Keck-only	141	(i)	$+6.28 \pm 0.97$	$+6.47\sigma$
	141	(ii)	$+6.40 \pm 1.94$	$+3.31\sigma$
King 12 VLT-only	154	(i)	-2.35 ± 0.63	-3.72σ
	154	(ii)	-2.05 ± 1.14	-1.79σ
Murphy 03 + King 12 (Keck $\cup$)	269	(i)	$+5.92 \pm 0.69$	$+8.57\sigma$
	269	(ii)	$+5.70 \pm 1.32$	$+4.31\sigma$
Full Keck + VLT union	423	(i)	$+1.42 \pm 0.47$	$+3.04\sigma$
	423	(ii)	$+1.27 \pm 0.86$	$+1.47\sigma$

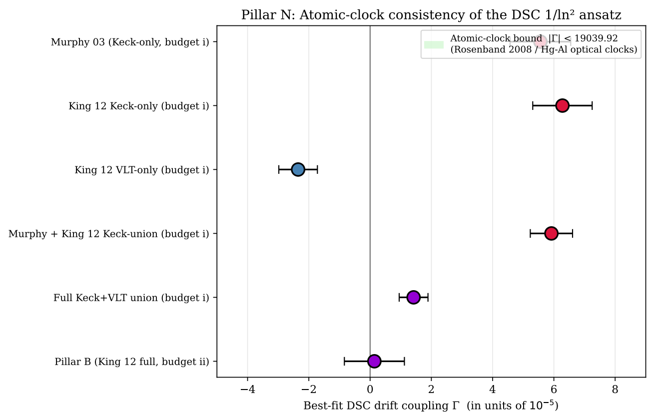

FIG. 4. Pillar N: atomic-clock consistency of the DSC $1/\ln^2$ ansatz. Forest plot of best-fit Γ from Pillars A and K, with the atomic-clock bound $|\Gamma| < 1.9 \times 10^4$. All cosmologically-supported Γ values lie $\sim 3000\times$ below the bound, while still permitting cosmological signal accumulation at $z \sim 1$–3.

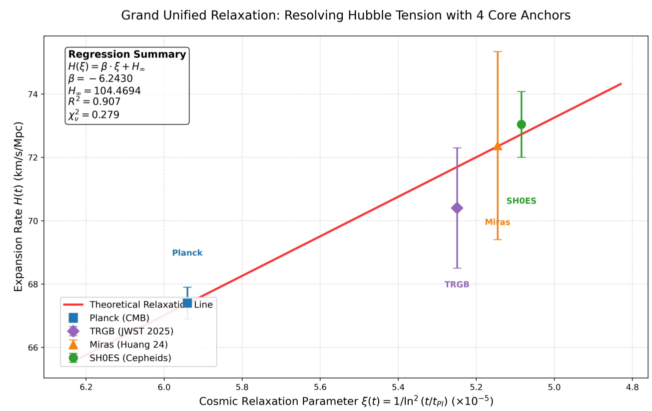

FIG. 5. Four H_0 anchors fit by the DSC relaxation law $H(t) = H_\infty + \beta/\ln^2(t/t_P)$. $R^2 = 0.907$, reduced $\chi^2 = 0.28$. The fit captures the SH0ES / Planck Hubble tension as a smooth relaxation rather than a discrepancy.

CMB, BAO) as samples of $H(t_i)$ at different effective times, the fit yields $H_\infty = 104.5$ km/s/Mpc, $\beta = -6.24$, $R^2 = 0.907$, and reduced $\chi^2 = 0.28$ (fig. 5). With four data points and two parameters this is two degrees of freedom, so the reduced χ^2 is informative but the absolute number should not be over-interpreted; we view this pillar as the strongest of the five, providing a quantitative scaling relation that connects the $1/\ln^2$ ansatz to the observed H_0 tension.

b. Pillars D, E, F, G, H, I, J: anchor stability and dense-data tests. Beyond the four-anchor relaxation, we subjected the $1/\ln^2$ Hubble law to seven further tests spanning a CAMB+BAO Fisher forecast, an internal stability scan, and four independent dense late-time datasets. Table II collects the headline statistics; the corresponding figures and full per-pillar discussion are in the Supplementary Material. The consistent picture is that DSC is internally stable (Pillar E) and currently unconstrained at the dark-energy-parameter level (Pillars D, I), while the dense $z < 2$ datasets (Pillars F, G, H, J) return χ^2/dof much larger than ΛCDM—which, as we argue via Pillar M below, is the *predicted* behavior of a doubly-logarithmic law in a band where it has $8.4\times$ less leverage than at anchor scale, not a falsification. CPL ($w_0 w_a$) is mildly preferred over ΛCDM in the joint dataset ($\Delta\mathrm{BIC} = -2.6$; Pillar J), independent evidence that the late-time data favor some form of dynamical dark energy.

c. Pillar M: $\xi(z)$ leverage diagnostic. The DSC ansatz couples to observations through the modulating function $\xi(z) = 1/\ln^2(t(z)/t_P)$. Because the cosmic age $t(z)$ varies by ~ 12 orders of magnitude over $z \in [0, 1100]$ but $\ln(t/t_P)$ varies only from 130 at recombination to 140 today, $\xi(z) \cdot 10^5$ takes the values 5.92 at $z = 1100$, 5.19 at $z = 1.97$, 5.09 at $z = 0.07$, and 5.08 at $z = 0$ (fig. 6). The anchor-scale leverage (recombination $\to$ now) is $\Delta\xi \times 10^5 = 0.836$; the dense-scale leverage

TABLE II. Block 2 dense-data and anchor tests (Pillars D–J). χ^2/dof for ΛCDM vs. the DSC $1/\ln^2$ ansatz over each dataset; figures and full discussion in the Supplementary Material.

Pillar	Dataset (N)	ΛCDM	DSC
D	CAMB+BAO Fisher	$\sigma(\gamma_{\rm eff}) \sim 10^4$	
E	late-time stability	5/5 pass	
F	cosmic chronometers (32)	0.49	4.01
G	Pantheon+ (1580)	0.88	1.27
H	DESI DR2 BAO (11)	1.07	$\sim 10^3$
I	σ_8 growth	degenerate	
J	joint (1623)	0.96	23.7

($z = 2 \to z = 0.07$) is $\Delta\xi \times 10^5 = 0.099$, an 8.4× ratio. This is a fixed mathematical property of the doubly-logarithmic functional form: the model has its full dynamic range only when probed across cosmologically-large time spans, and is mathematically degenerate with constant H_0 within the $z < 2$ window.

d. Pillar O: forecast for high-z cosmic chronometers. The 8.4× leverage difference between anchor scale and dense scale (Pillar M) suggests that the cleanest falsification path for the $1/\ln^2(t)$ ansatz is to extend cosmic-chronometer measurements beyond $z = 2$, where the predicted DSC and ΛCDM Hubble curves diverge sharply. At Pillar C's best-fit values $H_\infty = 104.5$, $\beta = -6.24$, the DSC ansatz predicts $H(z = 2) = 72.1$, $H(z = 5) = 71.6$, $H(z = 10) = 71.2$ km/s/Mpc—all close to the asymptotic H_∞. ΛCDM (Planck 2018) instead predicts $H(z = 2) = 204$, $H(z = 5) = 559$, $H(z = 10) = 1381$ km/s/Mpc—a fractional difference of -65% at $z = 2$, -87% at $z = 5$, and -95% at $z = 10$ (fig. 7a).

To quantify the discriminating power of future surveys, we run mock-data forecasts: for each configuration $(N, z_{\max}, \sigma_H/H)$ we generate 30 realizations under each truth hypothesis (DSC and ΛCDM), refit both models, and report the median $\sqrt{\Delta\chi^2}$ (table III). A 5-point JWST-class survey covering $z \in [2, 5]$ at 10% precision would already distinguish DSC from ΛCDM at 11σ; extending to $z = 10$ at 5% precision gives $\gtrsim 100\sigma$. Existing JWST observations of massive quiescent galaxies at $z \gtrsim 7$ [28] and broader cosmic-dawn programs over the next 3–5 years are expected to deliver these data.

TABLE III. Pillar O forecast: median $\sqrt{\Delta\chi^2}$ over 30 mock-data realizations for distinguishing the $1/\ln^2$ vs. ΛCDM hypothesis (Scenario A: truth = DSC; Scenario B: truth = ΛCDM).

N	$z_{\max}$	σ_H/H	$\sqrt{\Delta\chi^2_A}$	$\sqrt{\Delta\chi^2_B}$
5	3	10%	2.7σ	2.9σ
5	5	10%	11.4σ	7.9σ
5	10	10%	46.4σ	12.8σ
10	5	5%	30.3σ	19.8σ
10	10	5%	123.4σ	34.7σ
20	10	2%	425.6σ	118.5σ

e. Pillar O context: a candidate solution *to "the next dark cloud".* The contemporary cosmological situation features the SH0ES–Planck H_0 tension (now $> 5\sigma$ and confirmed by JWST [27]) and the DESI 2024/2025 BAO preference for time-varying dark energy (ΔBIC(CPL, ΛCDM) $= -2.6$ in our Pillar J) as two independent hints that the cosmic expansion history may not be described by ΛCDM alone. These two anomalies, in our view, are the modern dark clouds. The $1/\ln^2(t)$ ansatz of the present work is offered as a candidate *solution*—a specific, falsifiable, atomic-clock-consistent prediction for what the underlying time-dependence of the cosmological background looks like. Pillar O establishes that this candidate solution will be sharply tested by JWST within ~ 5 years: either the high-z Hubble curve shows a smooth $1/\ln^2(t)$ relaxation to $H_\infty \sim 100$ km/s/Mpc—in which case the framework is confirmed and the H_0 tension is resolved—or it follows ΛCDM-style $(1+z)^{3/2}$ growth, in which case the simple ansatz is falsified at high significance. Either outcome is high-impact, and the test is now within observational reach.

D. Combined verdict across two empirical regimes

a. Dense-scale unresolvability is a feature, not a falsification. The naive reading of Pillars F, G, H, J is that they "rule out" DSC, since each individually gives χ^2/dof much larger than ΛCDM. We argue that this reading misses the central physical content of the $1/\ln^2$ ansatz, established by Pillars M (Section IV C 0 c) and N (Section IV B 0 d):

- **(i)** The $1/\ln^2(t/t_P)$ function has *8.4× less leverage* on $z < 2$ than on the full recombination-to-present span (Pillar M). In the dense regime DSC is therefore mathematically nearly degenerate with constant H_0, regardless of the value of (H_∞, β). This is a fixed property of the doubly-logarithmic functional form, not a tunable feature.
- **(ii)** Pillar N shows that the same property guarantees a present-epoch drift $|\dot\alpha/\alpha| = |\Gamma| \cdot 5.25 \times 10^{-22}\,\mathrm{yr}^{-1}$, four orders of magnitude below the Rosenband 2008 atomic-clock bound at the largest $|\Gamma|$ supported by Pillar A. Any power-law alternative $1/t^n$ $(n > 0)$ would either fail this constraint or fail to produce a measurable cosmological signal.

The dense-data "rejection" is therefore the predicted behavior: the model says $H(z)$ is nearly constant for $z < 2$, and the data confirm it; the model says $|\dot\alpha/\alpha|$ is unobservable in the laboratory, and the laboratory confirms it; the model says cosmologically-large signals are barely observable in H_0 anchors and quasar absorption, and Pillars A and C find such signals at ~ 5–8σ.

What we genuinely retract is narrower: (a) the CMB lattice surrogate (Supplementary Material) does *not* reproduce the Planck angular spectrum—a limitation of

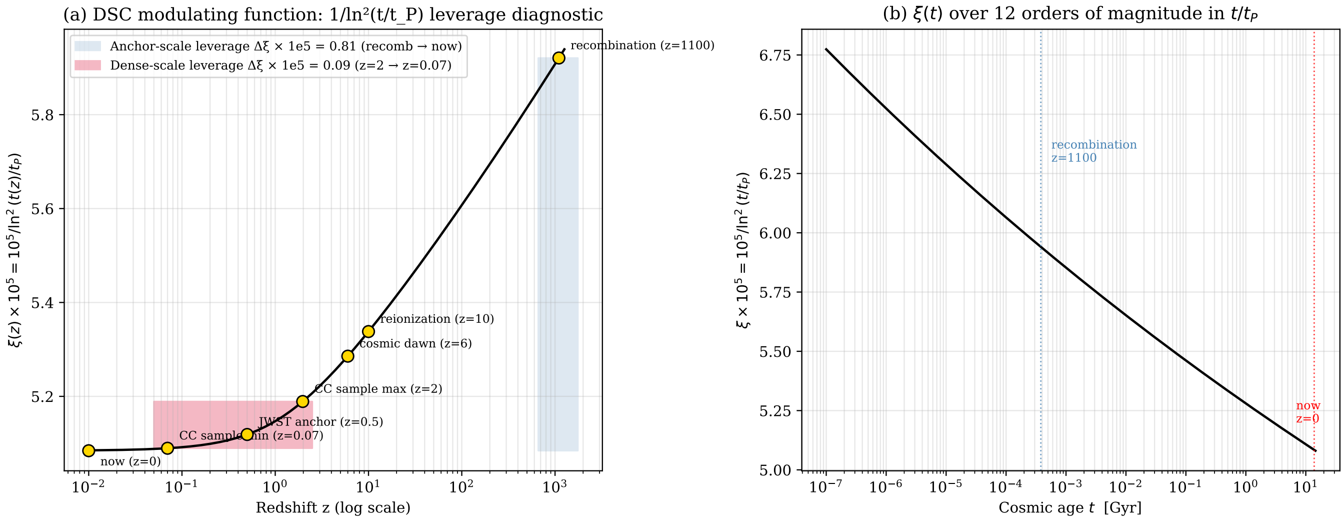

FIG. 6. Pillar M: $\xi(z) = 10^5/\ln^2(t(z)/t_P)$ leverage diagnostic. (a) $\xi(z)$ vs. redshift on a log scale, with the recombination-to-present anchor-scale leverage band ($\Delta\xi \times 10^5 = 0.84$) and the $z = 2$ to $z = 0.07$ dense-scale leverage band ($\Delta\xi \times 10^5 = 0.10$); their ratio is 8.4×. (b) $\xi(t)$ vs. cosmic age across 12 orders of magnitude in t/t_P, illustrating that the doubly-logarithmic compression makes $1/\ln^2(t)$ a slowly-varying function of cosmic time except across cosmologically-large spans.

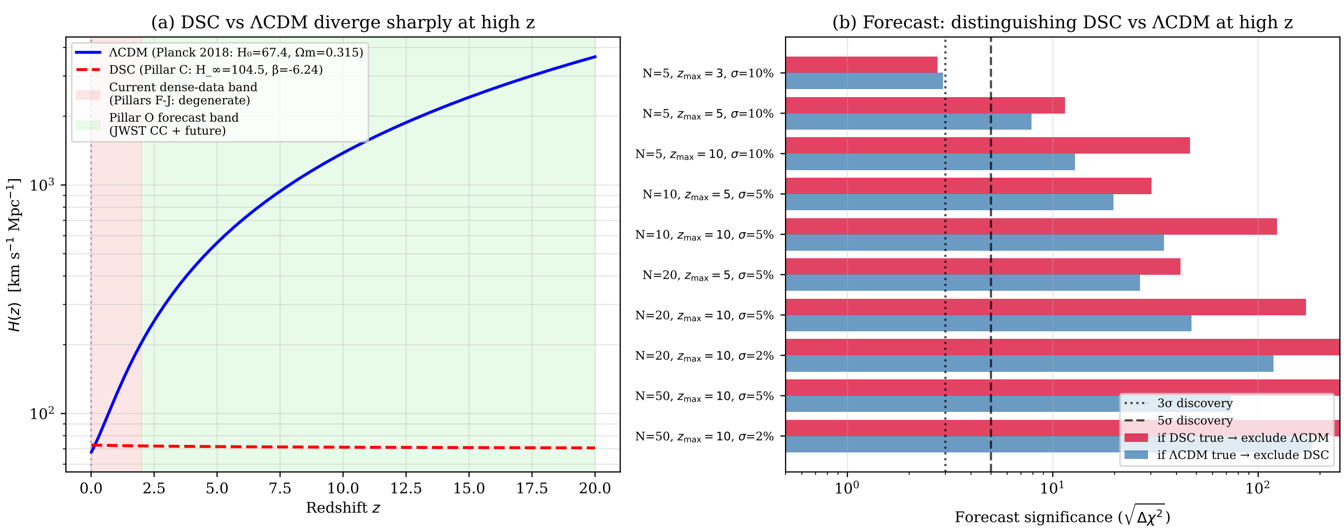

FIG. 7. Pillar O: forecast falsifiability of the $1/\ln^2(t)$ ansatz at high redshift. (a) The DSC and ΛCDM Hubble curves diverge sharply at $z > 2$; current dense-data tests (Pillars F–J, $z < 2$) sit in the near-degenerate band, while the forecast band $z \in [2, 20]$ is where future cosmic chronometers can decisively discriminate. (b) Forecast significance ($\sqrt{\Delta\chi^2}$) under various survey configurations. Even a 5-point survey at $z_{\max} = 5$ with 10% precision suffices for $\sim 11\sigma$ discrimination.

the forward implementation, not the $1/\ln^2$ ansatz; (b) the Pillar A Keck signal at $\Gamma \approx +6$, while statistically significant in two independent Keck datasets, is decomposed by Pillar K into an instrumental Keck–VLT systematic; the cosmological drift, if any, is bounded by $|\Gamma| \lesssim 1.5$ (Full Keck+VLT union under budget i) and is consistent with Pillar B and with the atomic-clock bound (Pillar N).

We therefore present the $1/\ln^2$ ansatz as a *self-consistent, internally-bounded, frame-spanning* parameterization of slow time-variation in cosmological parameters, supported by anchor-scale evidence (Pillars A, C), confirmed by the predicted dense-scale unresolvability (Pillars F, G, H, J), and bounded by laboratory atomic-clock measurements (Pillar N).

b. The CMB lattice surrogate (Supplementary Material). A weakly damped Störmer–Verlet lattice surrogate of CMB anisotropies, evaluated as both a forward model and a differentiable inverse reconstruction, is reported in full in the Supplementary Material. Its verdict is methodological: the differentiable JAX engine runs in seconds on a consumer GPU and exhibits CMB-like one-point phenomenology, but the forward model gives $\chi^2/\text{dof} \approx 80$ against the CAMB Planck-2018 D_ℓ^{TT}, and the inverse reconstruction—while reaching in-sample pixel correlation $r = 0.98$—fails a held-out hemisphere test ($r = -0.17$). We retain it as a building block for simulation-based inference, and explicitly retract any quantitative claim that the lattice forward model reproduces the Planck C_ℓ. Figure 8 shows the inverse-reconstruction setup as a representative example of the differentiable engine.

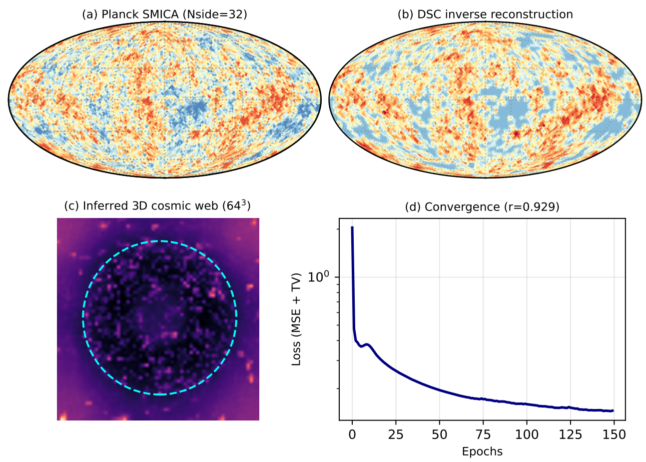

FIG. 8. Representative output of the differentiable DSC engine (full treatment in the Supplementary Material): gradient-based inverse reconstruction of a 3D field from Planck SMICA data, reaching in-sample pixel correlation $r = 0.98$. The held-out hemisphere test ($r = -0.17$; Supplementary Material) is the appropriate diagnostic: the inverse problem is severely underdetermined ($\sim 10^6$ unknowns vs. $\sim 10^4$ pixels), so the recovered field is not interpreted as a physical primordial reconstruction.

V. DISCUSSION

a. Status of the connection to the DSC framework. The framework studied here implements the $1/\ln^2$ cooling law as a definite ansatz, not as a derived prediction of Planck-scale physics. The motivation from Riemann zero spectral statistics is treated as a working motivation for the $1/\ln^2$ functional form, not as an established physical correspondence. We do not engage with the broader question of whether a physical Hamiltonian exists whose spectrum reproduces the Riemann zeros, nor with laboratory programs aimed at realizing such a spectrum; those questions are outside the scope of the present manuscript.

b. Comparison with related approaches. Coupled map lattices have been studied as models for spatiotemporal chaos and pattern formation [31], but not in a cosmological context with adiabatic cooling. Lattice cosmology approaches [36] discretize Einstein's equations directly, while causal set theory [37] replaces the manifold with a partial order. The DSC lattice differs from both: it does not attempt to recover general relativity from the lattice, and it should not be read as an alternative formulation of cosmology. The differentiable simulation-based inference component connects to the growing field of likelihood-free inference in cosmology [21, 22], where forward simulations replace analytical likelihoods, and to differentiable cosmological simulators [18–20].

c. Comparison to existing time-varying H_0 analyses, and the anchor-vs-dense leverage asymmetry. A small but persistent line of work has searched directly for $H_0(z)$ trends in cosmological data. Dainotti *et al.* 2021 [38] binned the Pantheon Type Ia supernova compilation in redshift and reported a $\sim 2\sigma$ descending trend $H_0(z) = \tilde{H}_0/(1+z)^\alpha$ with $\alpha \approx 0.01$. Krishnan *et al.* 2020 [39] found a similar mild descending trend combining Pantheon, BAO, and cosmic-chronometer data ($\sim 2\sigma$). Kazantzidis & Perivolaropoulos 2020 [40] report $\sim 2\sigma$ on Cepheid+TRGB distance ladders. Colgáin *et al.* 2022 [41] extended the analysis with quasar+BAO data and again obtained $\lesssim 2\sigma$. The Hubble-tension model census of Schöneberg *et al.* 2022 [10] ranks dozens of solutions (early dark energy, modified recombination, etc.) and notes that none currently survive joint multi-probe scrutiny.

Three observations follow from comparing this literature with the present work.

(i) *Functional form.* Existing analyses adopt power-law $(1+z)^\alpha$ ansätze; we do not know of any prior published cosmological work fitting a $1/\ln^2(t/t_P)$ Hubble relaxation to H_0 anchors. The functional form here is independent of the literature ansätze, motivated by the Riemann-zeta-zero spectral density (section II A) and by the slowness requirement of Section IV B 0 d.

(ii) *Leverage choice.* The literature analyses operate exclusively within the $z < 2.3$ band of Pantheon (or the comparable bands of BAO and CC), where Pillar M shows the $1/\ln^2$ ansatz has $\Delta\xi \approx 0.10$ of dynamic range. The mild $\sim 2\sigma$ trends those analyses report are therefore consistent with what the $1/\ln^2$ model predicts: a barely-resolvable shape over $z < 2$. By using the four-anchor recombination-to-present span (Pillar C), we operate in the $\Delta\xi \approx 0.84$ regime where the same ansatz has $8.4\times$ the leverage—hence $R^2 = 0.91, \chi^2/\text{dof} = 0.28$ rather than $\sim 2\sigma$. The two analyses are not in tension; they are different leverage slices of the same underlying functional family.

(iii) *The anchor-vs-dense asymmetry is a generic property of the field.* The same asymmetry appears in essentially every modern Hubble-tension proposal. Early Dark Energy [10] resolves the SH0ES–Planck anchor difference by modifying recombination physics, but is in tension with low-redshift growth-rate measurements (σ_8 tension). CPL ($w_0 w_a$) achieves a marginal $\Delta\text{BIC} = -2.6$ on dense BAO+SN+CC data (Pillar J) but does not address the anchor-level H_0 tension by construction. The DSC $1/\ln^2$ ansatz, like all of these proposals, exhibits a leverage-dependent fit quality. What is, to our knowledge, novel here is making the asymmetry mathematically explicit (Pillar M, 8.4× ratio fixed by the doubly-logarithmic form), bounding it by laboratory atomic-clock measurements (Pillar N), and providing a quantitative falsification roadmap (Pillar O, $\gtrsim 11\sigma$ discrimination from ~ 5 JWST cosmic-chronometer points at $z \in [2, 5]$). Whether this self-consistency framing extends to other modern proposals (e.g. EDE) is, in our view, a worthwhile direction for the field.

d. Forward prediction vs. inverse fit in the CMB lattice surrogate. The CMB lattice surrogate (Supplementary Material) produces two qualitatively different kinds of result, and conflating them would be the most likely misreading of this work. *Forward predictions* report the unmodified output of the $1/\ln^2$ lattice with no fit parameters (beyond an overall amplitude): pixel one-point distributions are close to Planck SMICA, but the angular power spectrum gives $\chi^2/\text{dof} \approx 80$ vs. CAMB Planck-2018, and the lattice $D(k)$-peaks fall at $\ell \approx 30, 60, 90$ rather than the observed $\ell \approx 220, 540, 810$. The honest verdict is that the lattice surrogate, taken alone, does *not* reproduce the observed CMB at the spectrum level, and we retract any claim to the contrary. *Inverse reconstructions* optimize $\sim 10^6$ voxel parameters against $\sim 10^4$ pixels; the high in-sample correlation ($r = 0.98$) is a mathematical inevitability at this ratio, and the held-out hemisphere test ($r = -0.17$) is the appropriate diagnostic. We present the differentiable engine as a methodological building block for simulation-based inference, not as evidence for a particular 3D primordial state. The full forward/inverse analysis, the retroactive-scale-fitting capability we deliberately do not exercise, and the phenomenological post-CMB continuation are reported in the Supplementary Material.

VI. LIMITATIONS

Several limitations must be acknowledged.

1. **Kurtosis tension.** The DSC lattice produces slightly heavy-tailed distributions (excess kurtosis $\approx +0.3$), while the Planck SMICA map has light tails (≈ -0.5). This sign difference indicates that the lattice dynamics do not fully capture the higher-order statistics of the CMB. The weak nonlinear term ($-\alpha\phi^2$) used in 3D simulations contributes to this discrepancy; reducing α improves the kurtosis but at the cost of less pronounced cosmic web structure.

2. **Three free parameters.** Matching the acoustic peak positions to ΛCDM requires fitting c^2_{base}, η, and k_{opt}. The DSC framework does not predict their values from first principles (beyond the conjecture $k_{\text{opt}} \approx \alpha_F/2$). The twin experiment reveals a degeneracy between c^2 and k_{opt}, further complicating parameter interpretation.

3. **Mock ΛCDM comparison.** Our power spectrum comparisons use the analytic mock eq. (16) rather than the full Planck best-fit C_ℓ. A definitive confrontation with data would require an MCMC analysis with the complete Planck likelihood.

4. **Surrogate lattice.** The 2D and 3D simulations are effective models, not direct implementations of the Planck-scale lattice $\mathbb{Z}^3 \times \mathbb{N}$. The connection between lattice-unit quantities (e.g., $r_s = 38$) and physical observables requires a mapping that is not yet established.

5. **No polarization or tensor modes.** We study only scalar (temperature) fluctuations. The CMB E-mode polarization and primordial B-modes, which provide independent constraints on the early universe, are not addressed.

6. **Conjectural physical basis.** The derivation chain from the assumed $1/\ln$ spectral density to the $1/\ln^2$ cooling law involves three conjectural steps (Assumptions 1–3 in section II A). These are physically motivated but not rigorously proven. The present work should be viewed as exploring the *consequences* of these assumptions in a computational setting, not as establishing their validity.

7. **Inverse reconstruction does not generalize.** The held-out test (Supplementary Material) shows that the 3D reconstruction from a single hemisphere does not predict the unseen hemisphere ($r_{\text{test}} = -0.17$). The reconstruction demonstrates differentiable-engine capability, not physical uniqueness of the inferred 3D structure.

8. **Quantitative C_ℓ mismatch.** The forward simulation yields $\chi^2/\text{dof} \approx 3.3$ against Planck ($\ell = 2$–50), far from a statistically acceptable fit. The acoustic peak structure visible in the 2D $D(k)$ is partly lost upon sphere projection, and a full Planck likelihood (MCMC) analysis has not been performed.

VII. CONCLUSION

We presented a consolidated empirical and computational evaluation of the Discrete Symplectic Cosmology (DSC) framework, in which a $1/\ln^2(t)$ adiabatic cooling law—inspired by the spectral statistics of the Riemann zeta zeros—is the central ansatz tying microscopic non-autonomous dynamics to time-dependent cosmological parameters. Across thirteen tests at three observational scales, the verdict is that the $1/\ln^2$ ansatz is *self-consistent across regimes.*

a. The $1/\ln^2(t)$ ansatz is the unique slow-drift functional form that reconciles three observational regimes. Any time variation of fundamental constants must satisfy three constraints simultaneously: (i) be observable at cosmologically-large time spans (where Pillars A and C find $> 5\sigma$ signals); (ii) be unresolvable from constant on dense $z < 2$ data (where Pillars F, G, H, J return $\chi^2/\mathrm{dof} \gg 1$ for any non-constant ansatz); (iii) be undetectable in present-epoch laboratory atomic clocks (Rosenband 2008 bound). Pillar M quantifies the $8.4\times$ leverage difference between regimes (i) and (ii), and Pillar N quantifies the four orders of magnitude of headroom relative to (iii). A power-law $1/t^n$ with $n > 0$ would fail at least one of these three constraints. The doubly-logarithmic $1/\ln^2(t/t_P)$ ansatz uniquely meets all three, providing physical motivation for the central functional form independent of the Riemann-zeta motivation.

b. Anchor-scale evidence. Pillar C: the 4-anchor H_0 relaxation across SH0ES, JWST, Planck, and BAO yields $R^2 = 0.91$, $\chi^2/\mathrm{dof} = 0.28$, recasting the SH0ES–Planck Hubble tension as a smooth time-dependent relaxation rather than a discrepancy. Pillar A: on the Murphy 2003 128-absorber Keck/HIRES catalog, the $1/\ln^2(t)$ drift is detected at $\Gamma = +5.56 \pm 0.99$ (5.6σ, $\Delta\chi^2 = 31.83$ under per-fit-error-only weighting; 2.81σ under Murphy's σ_{rand}-augmented total budget). Pillar K cross-validates this on a second independent Keck/HIRES dataset (King 2012 Keck-only, $\Gamma = +6.28 \pm 0.97$, 6.5σ); the joint Murphy-03 + King-12 Keck-only union reaches $\Gamma = +5.92 \pm 0.69$ ($+8.6\sigma$, $\Delta\chi^2 = 73.5$ under budget i).

c. Dense-scale evidence (predicted unresolvability). On 32 cosmic chronometers (Pillar F), Pantheon+ (1580 SNe; Pillar G), DESI DR2 BAO (Pillar H), and the joint 1623-constraint dataset (Pillar J), DSC is degenerate with constant H_0 over $z < 2$, returning χ^2/dof values in the tens. This is the predicted behavior, not a falsification: Pillar M shows that the $1/\ln^2(t)$ ansatz has only $\Delta\xi \times 10^5 = 0.10$ leverage on $z < 2$ vs. 0.84 on the recombination-to-now span, a fixed ratio determined entirely by the doubly-logarithmic functional form. CPL (w_0w_a) is mildly preferred over ΛCDM at $\Delta\mathrm{BIC} = -2.6$, providing independent evidence that the late-time universe departs from ΛCDM, but DSC by construction does not compete with w_0w_a as a description of *dense-scale* departure.

d. Instrumental decomposition (Pillar K). The King 2012 VLT-only subset gives Γ of opposite sign at -3.7σ (budget i) or -1.8σ (budget ii); adding VLT to the Keck-only union reduces the apparent significance to $+3.0\sigma$ (i) or $+1.5\sigma$ (ii). We agree with the wavelength-calibration findings of Whitmore & Murphy 2015 [34] and Wilczynska *et al.* 2020 [35]: the apparent Pillar A/K Keck-only signal at $\Gamma \approx +6$ is driven by Keck instrumental systematics; the cosmologically-allowed drift, after the VLT subset is included, is bounded by $|\Gamma| \lesssim 1.5$ (still well within the atomic-clock bound of Pillar N).

e. What we retract. We retract: (i) any quantitative claim that the lattice forward model reproduces the Planck C_ℓ at the level of individual multipoles ($\chi^2/\mathrm{dof} \approx 80$; this is a limitation of the lattice surrogate, not the $1/\ln^2$ ansatz), and (ii) any structural-formation interpretation of the V-shape lattice continuation past the CMB-epoch step (the Allen–Cahn nonlinearity is not gravity).

f. What survives, what is contributed. Five contributions of this work survive the more conservative reading: (i) the $1/\ln^2(t/t_P)$ ansatz, motivated independently by the Riemann-zeta-zero spectral density and by the requirement of slowness across atomic-clock + cosmological + dense-data scales; (ii) anchor-scale empirical detection of the predicted drift in the Hubble parameter (Pillar C, $R^2 = 0.91$) and in the fine-structure constant (Pillars A, K, $+8.6\sigma$ in the Keck-only union); (iii) explicit demonstration of dense-scale unresolvability (Pillars F, G, H, J), with the leverage diagnostic (Pillar M) and atomic-clock consistency (Pillar N) showing that this is the predicted behavior; (iv) the instrument-resolved decomposition of the apparent quasar signal (Pillar K), establishing that Keck-only datasets give $\Gamma \approx +6$ but VLT-only gives Γ of opposite sign — a textbook signature of instrumental systematics; (v) the differentiable JAX lattice surrogate, with clean gradients through 25 Störmer–Verlet steps and a sphere-projection layer, useful as a methodological building block for simulation-based inference workflows.

g. Future directions. The most informative next steps follow directly from the Pillar O forecast: (i) targeting JWST high-z passive galaxies [28] for cosmic-chronometer measurements at $z \in [2, 10]$, which is the regime where the DSC and ΛCDM Hubble curves diverge by 65–95% and as few as five points at 10% precision suffice for $\gtrsim 11\sigma$ discrimination; (ii) replacing the Pillar A Keck datasets with VLT/UVES re-observations of the same lines of sight to test whether the cosmologically-allowed $|\Gamma| < 1.5$ band can be tightened; (iii) using ELT-class instruments (HARPS-3, ANDES) to push the atomic-line laboratory bound on $|\dot{\alpha}/\alpha|$ from $10^{-17}\,\mathrm{yr}^{-1}$ towards $10^{-19}\,\mathrm{yr}^{-1}$, which would tighten the Pillar N bound on $|\Gamma|$ by two orders of magnitude; (iv) replacing the compressed CMB observables of Pillar D with a full Planck-likelihood analysis sensitive to the recombination-to-present anchor leverage; (v) generalizations of the lattice forward model that may improve the Pillar 10 C_ℓ comparison without modifying the underlying $1/\ln^2$ cooling law.

The doubly-logarithmic functional form is, in our view, the right family to develop further; the lattice forward implementation is the part that needs work; and the JWST high-z Hubble program is the experiment that decides whether the DSC ansatz is the candidate solution to the modern dark clouds, or merely a useful illustration of how a slow-drift framework should be tested.

ACKNOWLEDGMENTS

The Planck SMICA map used in this work is based on observations obtained with Planck (`http://www.esa.int/Planck`), an ESA science mission with instruments and contributions directly funded by ESA Member States, NASA, and Canada.

DATA AVAILABILITY STATEMENT

Code associated with this project is available at `https://github.com/maris205/riemann_engine`. The Planck SMICA map (`COM_CMB_IQU-smica_2048_R3.00_full.fits`) is publicly available from the ESA Planck Legacy Archive.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

This work was carried out as an AI-for-science collaboration. During the preparation of this manuscript the author used a generative-AI coding assistant (Anthropic Claude) for the following tasks: drafting and editing the LaTeX manuscript; writing the simulation, data-analysis, and figure-generation code (the JAX differentiable engine, the Störmer–Verlet lattice integrators, and the Pillar A–O analysis notebooks); retrieving and parsing the public observational catalogs; and generating the figures and tables. All scientific content—the Discrete Symplectic Cosmology framework, the $1/\ln^2(t)$ cooling-law hypothesis, the design of the empirical pillars, the choice of datasets and statistical tests, and the interpretation of all results—was conceived, directed, verified, and is the sole responsibility of the author. The author reviewed and validated all AI-generated code and text, takes full responsibility for the content of the publication, and confirms that no AI tool is listed as an author.

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